

Highlights

A boundary-integrated closure for renormalised meshless operators on elliptic boundary value problems

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- A unified boundary closure is introduced, integrating Neumann and Dirichlet or Robin conditions into renormalised meshless operators on a single code path.
- The method detects the free surface from the point cloud and enforces it without tuning parameters, as the enforcement strength cancels to the total kernel weight.
- A general Gram projector, applied in physical space before renormalisation, treats non-orthogonal corners exactly.
- An algebraic-ghost closure restores Neumann accuracy without extra unknowns, approaching second-order convergence under the sum normalisation.
- Consistency is demonstrated in a Helmholtz-Hodge decomposition, where the residual divergence represents the non-adjointness defect of the operators and vanishes under refinement.

A boundary-integrated closure for renormalised meshless operators on elliptic boundary value problems

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Abstract

A boundary-condition closure for renormalised, least-squares corrected meshless Laplacian operators is presented. The closure is algebraic and operates in strong form. Unlike other methods, the approach does not pad the domain with auxiliary particles, represent the boundary with a level set, or enforce the conditions weakly. Instead, a single code path folds both Neumann and free-surface conditions directly into the discrete collocation row. The algorithm detects the boundary from the point cloud, requiring no tuning parameters, and organises the closure by the type of condition. A Neumann flux goes to the right-hand side, while a Dirichlet or Robin value enters the diagonal. The Neumann flux is imposed using a tangential projection in physical space before renormalisation. A general Gram projector treats non-orthogonal corners exactly. Furthermore, a complementary algebraic-ghost closure completes the truncated boundary stencil by reflecting neighbours and the node itself across the local reflection group, requiring no extra unknowns. The free surface is detected from the support-deficiency geometry of the cloud. Its enforcement strength cancels out to the total kernel weight, removing any penalty or threshold parameters. The proposed method, named

FIRM (Flux-Integrated Renormalised Meshless), is verified against analytical ground truths and irregular-domain Poisson benchmarks. Both regular and highly irregular point clouds in two dimensions are tested against a second-order generalised finite difference baseline, demonstrating that the operators remain exact for linear fields on any cloud. The value closure supraconverges, and the algebraic-ghost closure reduces the Neumann-limited error by about an order of magnitude. With the sum normalisation, the method approaches second-order convergence at straight boundaries. Finally, the renormalised operators are shown to be consistent for a single Helmholtz-Hodge decomposition. The residual divergence equals the non-adjointness defect of the operators, which vanishes under refinement.

Keywords: meshless method, boundary conditions, renormalised Laplacian, free surface, Neumann, Poisson equation

1. Introduction

Solving elliptic boundary value problems on irregular domains is a common requirement in computational physics. This task becomes particularly difficult when using meshless discretisations. In these methods, the computational nodes move freely, meaning no conforming mesh represents the boundary. This challenge is observed in the pressure projection step of incompressible free-surface flows. It also appears in diffusion processes, electrostatics, and seepage through porous media on irregular or evolving domains. A companion study introduced a class of renormalised meshless Laplacian operators for the Poisson and diffusion equations [1]. These operators use a local least-squares moment matrix for correction. Consequently, the operators

maintain consistency even on highly irregular point arrangements, where the classical SPH Laplacian fails to converge [2, 3, 4]. This least-squares correction belongs to the kernel-gradient renormalisation lineage, which researchers originally developed to restore particle consistency [5, 6, 7]. However, that previous study only focused on the interior operator. That study reported Dirichlet and Robin results on prescribed domains, leaving several key questions open. Specifically, how are boundary conditions integrated when the boundary is detected from the point cloud rather than prescribed? How can Neumann and free-surface boundaries be handled on a single code path? The present work closes this boundary gap. The proposed approach complements and is orthogonal to the work of Asai et al. [8], who recently extended the same interior operator to full second-order consistency on disordered clouds.

Researchers have proposed several methods to handle boundaries on meshless or irregular domains. These methods differ less in their final accuracy and more in what is required from the geometry. In the first approach, a background grid represents and defines the boundary geometrically. For instance, in the level-set tradition, researchers built second-order-accurate symmetric discretisations of the Poisson equation on irregular domains using a ghost-value treatment [9, 10, 11, 12]. Subsequent works extended this treatment to Robin conditions on implicit interfaces [13] and to Voronoi interface constructions [14]. This approach requires an adaptive Cartesian grid and a level-set description of the geometry. In a second approach, auxiliary material represents the boundary. In SPH and ISPH, dummy or mirror particles, repulsive forces, or generalised wall conditions carry solid walls and free surfaces [15, 16, 17]. Other variants use semi-analytical bound-

ary integrals [18, 19, 20], or corrected Laplacians from the moving-particle semi-implicit category [21, 22, 23, 24, 25]. While effective, these treatments typically introduce auxiliary particles and require tuning a length scale or a threshold. For example, a thresholded density or kernel-sum criterion usually identifies the free surface. In a third approach, the formulation appends the boundary condition as additional algebraic rows. In the generalised finite difference method (GFDM) and the finite-pointset method (FPM), weighted least squares fits a full Taylor basis, and the boundary conditions enter as constraint rows [26, 27, 28, 29]. Similarly, radial-basis-function finite differences recover boundary accuracy by adding ghost nodes outside the domain [30, 31, 32]. Across all these techniques, boundary enforcement comes at a price. It demands stored degrees of freedom, a background grid, or a tuning parameter.

A separate line of work imposes the boundary condition in algebraic form, rather than extending the domain with extra geometry. Here, we describe three main categories. First, the formulation incorporates the condition directly into the trial space from the start, so that every valid field satisfies it by construction. For example, the R-function method builds boundary-conforming solution structures from implicit geometry [33, 34]. In weighted B-splines, a distance function to the boundary weights a B-spline basis [35]. This same concept underlies the exact imposition of boundary conditions in neural and physics-informed solvers using distance fields [36, 37]. In the second category, the formulation injects the boundary condition into the variational form instead of the basis. In methods like element-free Galerkin, the moving-least-squares shape functions lack the Kronecker-delta property.

This lack prevents pinning nodal values directly [38, 39]. Consequently, researchers enforce essential conditions weakly through Nitsche’s method [40] or a penalty method [41], though strong-type variants have also appeared recently [42]. In the third category, one does not prescribe a boundary condition at all. Instead, correcting the truncated kernel summation near the boundary keeps the operator consistent. This includes Shepard normalisation [43], corrective or reproducing-kernel formulations [44], and the boundary truncation-error analysis of Quinlan et al. [45]. This third category is the closest ancestor to the renormalised operators, though the method restores consistency without prescribing the actual boundary value or flux. The proposed closure is algebraic in a similar sense, but it differs from all three categories. The boundary condition is not incorporated into the trial space, nor is background integration in a variational form required. Instead, the condition is integrated directly into a strong-form collocation row. The type of the condition organises this row. A Neumann flux goes to the right-hand side, while a Dirichlet or Robin value enters the diagonal of the same row. Furthermore, the algorithm detects the boundary from the point cloud, requiring no tuning parameters.

The main contribution of this work is a boundary closure for the renormalised operators of [1]. We provide three properties that are not found conjoined in existing methods. First, it is unified. A single kernel assembles the interior, the Neumann boundary, the free surface, and the contact lines on one code path. Only the type of boundary condition distinguishes these cases. A Neumann flux enters the right-hand side, and a Dirichlet or Robin value enters the diagonal of the same collocation row. Second, it is

geometry-detected. The free surface is identified from the support-deficiency geometry of the cloud, requiring no surface particles and avoiding level sets. Third, it is parameter-free. The enforcement strength cancels out to the total kernel weight, which is a purely geometric quantity. A parameter-free activation function leaves the interior nodes untouched. Thus, no penalty parameters, thresholds, or tuning lengths are introduced. A complementary algebraic-ghost closure is also introduced. This closure resolves the accuracy bottleneck of the flux-only Neumann treatment. It does so by completing the truncated boundary stencil using reflected samples. The algorithm forms these samples on the fly and never stores them globally, adding no extra unknowns. As a smaller supporting result, a single-sum interior operator, termed the smooth-naive operator, is presented. It bridges the naive and sum versions of [1] using a trace-based normalisation. It preserves the corrected summation of the sum version while using a single neighbour loop. The scope is restricted to the linear elliptic Poisson problem. Applications and extensions are discussed in the conclusions.

2. Meshless operators

In this section, we describe the three meshless second-derivative operators used in this study. We adopt a unified notation. The first is the renormalised differencing operator of [1], which we abbreviate as LDD, after the Lagrangian differencing dynamics framework [46]. We build the new boundary closure in Section 3 directly on this operator. The second is a generalised finite difference (GFDM) operator, which serves as a second-order baseline. The third is the second-order renormalised operator of [8], which

we re-express in a kernel-gradient-free form to match the present framework. Consider a point cloud Ω with nodal spacing Δ . Each node i interacts with neighbours j within a compact support radius $h = \kappa\Delta$. A kernel function $W_{ij} = W(|\mathbf{x}_{ij}|, h)$ governs this interaction, where we define the relative offset as $\mathbf{x}_{ij} = \mathbf{x}_j - \mathbf{x}_i$ and the field difference as $f_{ij} = f_j - f_i$. For all tests, we choose the poly6 function as the kernel:

2.1. Lagrangian Differencing Dynamics (LDD)

In the renormalised operators of [1], the formulation consumes and renormalises the raw kernel weights. This makes the operators kernel-agnostic in a sense we will make precise. We define several local quantities using the kernel: These are the support-deficiency vector $\mathbf{o}_i$, the total weight S_i , the second-moment renormalisation tensor $\mathbf{M}_i$, and its inverse $\mathbf{B}_i$:

$$\mathbf{o}_i = \sum_j W_{ij} \mathbf{x}_{ij}, \quad (1)$$

$$S_i = \sum_j W_{ij}, \quad (2)$$

$$\mathbf{M}_i = \sum_j W_{ij} \mathbf{x}_{ij} \otimes \mathbf{x}_{ij}, \quad (3)$$

$$\mathbf{B}_i = \mathbf{M}_i^{-1}. \quad (4)$$

The vector $\mathbf{o}_i$ vanishes when the support is perfectly symmetric. Near a boundary, it points inward. The inverse moment tensor $\mathbf{B}_i$ is symmetric and positive-definite, provided that $\mathbf{M}_i$ has full rank. This condition requires at least $d + 1$ non-degenerate neighbours. We write the renormalised gradient as:

$$\langle \nabla f \rangle_i = \mathbf{B}_i \sum_j W_{ij} f_{ij} \mathbf{x}_{ij}, \quad (5)$$

which remains exact for linear fields on any point cloud. We can easily verify this linear exactness. For a linear field $f = \mathbf{a} \cdot \mathbf{x} + b$, the difference is $f_{ij} = \mathbf{a} \cdot \mathbf{x}_{ij}$. Using the definition of $\mathbf{M}_i$ in Eq. (3), we evaluate the gradient:

$$\begin{aligned} \langle \nabla f \rangle_i &= \mathbf{B}_i \sum_j W_{ij} (\mathbf{a} \cdot \mathbf{x}_{ij}) \mathbf{x}_{ij} \\ &= \mathbf{B}_i \left(\sum_j W_{ij} \mathbf{x}_{ij} \otimes \mathbf{x}_{ij} \right) \mathbf{a} = \mathbf{B}_i \mathbf{M}_i \mathbf{a} = \mathbf{a}, \end{aligned} \quad (6)$$

Here, the renormalisation by $\mathbf{B}_i = \mathbf{M}_i^{-1}$ cancels out the unbalanced moments that a truncated support causes.

We obtain the Laplacian from a Taylor expansion. Subtracting the reconstructed gradient removes the first-order term. Two existing members of the renormalised family from [1] provide useful context. In the naive version, we weight each neighbour by the reciprocal squared distance:

$$\langle \nabla^2 f \rangle_i^{\text{naive}} = 2d \sum_j \frac{W_{ij}}{|\mathbf{x}_{ij}|^2} (f_{ij} - \mathbf{x}_{ij} \cdot \langle \nabla f \rangle_i), \quad (7)$$

This naive version works well on regular clouds. However, it exhibits high sensitivity to close neighbours. A single close pair can make the term $1/|\mathbf{x}_{ij}|^2$ extremely large, causing numerical issues. The sum version resolves this sensitivity by replacing individual weights with a single self-normalising ratio:

$$\langle \nabla^2 f \rangle_i^{\text{sum}} = 2d \frac{\sum_j W_{ij} w_{ij} f_{ij}}{\sum_j W_{ij} w_{ij} |\mathbf{x}_{ij}|^2}, \quad (8)$$

where we define the correction weight as $w_{ij} = 1 - \mathbf{x}_{ij} \cdot \mathbf{B}_i \mathbf{o}_i$. The denominator is the operator's own action on $\frac{1}{2}|\mathbf{x}|^2$, which makes it exact for the isotropic quadratic by construction.

We introduce a new operator, which we name the smooth-naive operator. We derive it from the naive form in Eq. (7) but use a smoothed normalisation.

Instead of the per-neighbour factor $2d/|\mathbf{x}_{ij}|^2$, we use a single tensor quantity. We build this quantity directly from the inverse moment tensor:

$$N_i = \frac{2}{d} \text{tr } \mathbf{B}_i. \quad (9)$$

The scaling factor $2/d$ comes from the spherical-mean identity:

$$\nabla^2 f = \lim_{r \rightarrow 0} \frac{2d}{r^2} (\bar{f}_r - f), \quad (10)$$

and yields the correct second-derivative magnitude in any dimension. A three-dimensional constant-Laplacian test in Section 4 confirms this scaling. Using this normalisation, we write the corrected second difference as:

$$\langle \nabla^2 f \rangle_i = N_i \sum_j W_{ij} (f_{ij} - \mathbf{x}_{ij} \cdot \langle \nabla f \rangle_i). \quad (11)$$

Note that $\langle \nabla f \rangle_i$ does not depend on the neighbour index j . Therefore, we can factor the support offset $\sum_j W_{ij} \mathbf{x}_{ij} = \mathbf{o}_i$ out of the second term. This simplification yields $\sum_j W_{ij} \mathbf{x}_{ij} \cdot \langle \nabla f \rangle_i = \mathbf{o}_i \cdot \langle \nabla f \rangle_i$. Substituting Eq. (5) and using the symmetry of $\mathbf{B}_i$ yields the following:

$$\begin{aligned} \mathbf{o}_i \cdot \langle \nabla f \rangle_i &= \mathbf{o}_i \cdot \mathbf{B}_i \sum_j W_{ij} f_{ij} \mathbf{x}_{ij} \\ &= \sum_j W_{ij} f_{ij} (\mathbf{x}_{ij} \cdot \mathbf{B}_i \mathbf{o}_i), \end{aligned} \quad (12)$$

Consequently, the double sum collapses, yielding a single-sum renormalised Laplacian:

$$\langle \nabla^2 f \rangle_i = N_i \sum_j W_{ij} f_{ij} (1 - \mathbf{x}_{ij} \cdot \mathbf{B}_i \mathbf{o}_i) \equiv N_i \sum_j W_{ij} f_{ij} w_{ij}. \quad (13)$$

Equation (13) bridges the two older formulations from [1]. Its summand $W_{ij} w_{ij} f_{ij}$ matches the numerator of the sum version in Eq. (8). Meanwhile,

its normalisation N_i behaves as a smoothed version of the naive weighting in Eq. (7). As a result, the smooth-naive operator retains the corrected, offset-shifted summation. It computes everything in a single neighbour loop, avoiding the need for a second summation pass. Furthermore, the three normalisations coincide when the support is isotropic. Under isotropic conditions, we obtain $w_{ij} \rightarrow 1$ and $\mathbf{M}_i = \frac{1}{d}(\sum_j W_{ij} |\mathbf{x}_{ij}|^2) \mathbf{I}$. This leads to the following relation:

$$N_i^{\text{den}} \equiv \frac{2d}{\sum_j W_{ij} w_{ij} |\mathbf{x}_{ij}|^2} \xrightarrow{\text{isotropic}} \frac{2d}{\sum_j W_{ij} |\mathbf{x}_{ij}|^2} = \frac{2}{d} \text{tr } \mathbf{B}_i = N_i, \quad (14)$$

Thus, the naive, sum, and smooth-naive operators all simplify to the same expression on a regular lattice. They differ only when support anisotropy is present. The denominator N_i^{den} in Eq. (14) normalises the sum operator of [1]. We retain this denominator normalisation as a more accurate alternative. Because it is exact for $|\mathbf{x}|^2$, the sum operator is about 8 to 15% more accurate than the smooth-naive version under particle disorder. This advantage grows as the chaos level increases. Specifically, the sum-to-smooth-naive error ratio falls from 1.00 on a regular lattice to 0.73 at 80% chaos, as we show in Section 5 and Fig. 16. We use the smooth-naive operator as the default because we avoid the denominator sum. The sum operator remains available as a more accurate refinement.

The correction weight makes Eq. (13) exact for linear fields on any point arrangement. This holds even for the truncated supports found at boundaries. To see why, let $f = \mathbf{a} \cdot \mathbf{x} + b$, which gives the difference $f_{ij} = \mathbf{a} \cdot \mathbf{x}_{ij}$.

We substitute this into the Laplacian equation:

$$\begin{aligned}\langle \nabla^2 f \rangle_i &= N_i \left[\sum_j W_{ij}(\mathbf{a} \cdot \mathbf{x}_{ij}) - \sum_j W_{ij}(\mathbf{a} \cdot \mathbf{x}_{ij})(\mathbf{x}_{ij} \cdot \mathbf{B}_i \mathbf{o}_i) \right] \\ &= N_i [\mathbf{a} \cdot \mathbf{o}_i - \mathbf{a} \cdot \mathbf{M}_i \mathbf{B}_i \mathbf{o}_i] = 0.\end{aligned}\tag{15}$$

Since $\mathbf{M}_i \mathbf{B}_i = \mathbf{I}$, the two terms cancel out. This cancellation occurs on any support, whether symmetric or truncated. We refer readers to [1, 47, 48] for a rigorous derivation and convergence analysis of these operators.

Equation (13) is linear-consistent. However, it is generally not second-order consistent on a disordered cloud. Its pointwise truncation error is $O(\Delta^2)$ on a regular lattice. Yet, under fixed-relative jitter, this error becomes $O(1)$ and does not vanish as $\Delta \rightarrow 0$. In spite of this, the global Poisson solution still converges. We call this phenomenon supraconvergence. It occurs because the $O(1)$ pointwise error is the discrete divergence of a consistent flux, which the global system solve effectively damps out. [1] established this behaviour. It serves as the framework for the accuracy results, and we revisit it in Section 3.6.

2.2. Generalised finite differences (GFDM)

We use a generalised finite difference operator of the finite-pointset type as a second-order reference [26, 27, 28]. At each node, we construct a full quadratic Taylor basis on the neighbour offsets:

$$\mathbf{p}_{ij} = [x_{ij}, y_{ij}, \tfrac{1}{2}x_{ij}^2, x_{ij}y_{ij}, \tfrac{1}{2}y_{ij}^2]^\top, \tag{16}$$

We fit this basis to the field differences f_{ij} using weighted least squares with the kernel weights W_{ij} . From this fit, we read off the Laplacian as the sum

of the two pure second-derivative coefficients:

$$\langle \nabla^2 f \rangle_i^{\text{GFDM}} = \mathbf{e}^\top (\mathbf{P}^\top \mathbf{W} \mathbf{P})^{-1} \mathbf{P}^\top \mathbf{W} \mathbf{f}, \quad (17)$$

where we define the selection vector as $\mathbf{e} = [0, 0, 1, 0, 1]^\top$. Here, $\mathbf{P}$ is the matrix of basis rows $\mathbf{p}_{ij}^\top$, $\mathbf{W} = \text{diag}(W_{ij})$, and $\mathbf{f} = (f_{ij})_j$. This operator is exact for quadratics. It is therefore second-order consistent in the interior on both regular and disordered clouds. However, this comes at the cost of solving a 5×5 local system in two dimensions. It also requires at least five well-spaced neighbours. On a regular grid, it simplifies to the standard five-point stencil. We impose the Neumann flux as an exact equality constraint using a local Karush-Kuhn-Tucker saddle system, following the finite-pointset constraint approach [27, 28]. We also test a soft penalty-flux variant, which we refer to as the penalty form, and show in Section 5.2 that it fails to converge.

2.3. Second-order operators in LDD form

The cross-term free LDD operator described in Section 2.1 is linear-consistent. Asai et al. [8] proposed a systematic taxonomy of renormalised second-derivative models. In that work, they identify a Block Diagonal operator. This is equivalent to the corrected Laplacian of Schwaiger [49] and to the present family. The Naive variant, on the other hand, coincides with the Taylor-series form of Lian et al. [50]. To recover full second-order pointwise consistency under particle disorder, we must use the Full Inverse operator. In this operator, we reinstate the cross-derivative coupling that the block-diagonal reduction dropped. This is similar to other consistent meshless second-derivative models, such as those by Español and Revenga [51], the least-squares moving-particle method of Tamai and Koshizuka [52], and the

GFDM operator in Section 2.2. Unfortunately, the terminology in this area is not uniform. What Asai et al. [8] call Block Diagonal is actually the variant that the LDD literature terms Full Inverse [1, 46].

This operator achieves second-order accuracy by fitting the full set of second derivatives, including the cross term, rather than just the trace. However, we never differentiate the kernel in this framework. The gradient in Eq. (5) and the Laplacian in Eq. (13) rely only on the scalar weight W_{ij} and the renormalisation matrix $\mathbf{B}_i$. Therefore, we adapt this second-order concept to a kernel-gradient-free form. We express it as a single weighted least-squares fit using the same renormalise-by- $\mathbf{B}_i$ approach. We collect the second derivatives in a vector $\mathbf{u}_i = [\partial_{xx}^2 f, \partial_{yy}^2 f, 2\partial_{xy}^2 f]^\top$. We also collect the quadratic offsets in the basis $\mathbf{s}_{ij} = [x_{ij}^2, y_{ij}^2, x_{ij}y_{ij}]^\top$. We represent the renormalised gradient in Eq. (5) with the stencil $\langle \nabla f \rangle_i = \sum_j \mathbf{c}_{ij} f_{ij}$, where $\mathbf{c}_{ij} = W_{ij} \mathbf{B}_i \mathbf{x}_{ij}$. Subtracting this gradient from the second difference leaves a second-order remainder:

$$g_{ij} \equiv f_{ij} - \mathbf{x}_{ij} \cdot \langle \nabla f \rangle_i = \frac{1}{2} \hat{\mathbf{s}}_{ij} \cdot \mathbf{u}_i. \quad (18)$$

In this equation, we must correct the basis for the second-derivative content that the renormalised gradient itself carries:

$$\hat{\mathbf{s}}_{ij} = \mathbf{s}_{ij} - \mathbf{C}_i \mathbf{x}_{ij}, \quad (19)$$

where we define the moment-offset correction matrix as:

$$\mathbf{C}_i = \sum_j W_{ij} \mathbf{s}_{ij} (\mathbf{B}_i \mathbf{x}_{ij})^\top. \quad (20)$$

The term $\mathbf{C}_i \mathbf{x}_{ij}$ acts as the kernel-gradient-free counterpart to the $R_{(2)}$ correction of [8]. If we omit this term, the fit collapses back to the linear-consistent

class, and we lose second-order accuracy under disorder. We solve the coefficients from the weighted normal equations:

$$\mathbf{M}_i^{\text{FI}} \mathbf{u}_i = 2 \sum_j W_{ij} \hat{\mathbf{s}}_{ij} g_{ij}, \quad (21)$$

where we define the second-order moment matrix as $\mathbf{M}_i^{\text{FI}} = \sum_j W_{ij} \hat{\mathbf{s}}_{ij} \otimes \hat{\mathbf{s}}_{ij}$. This is a small 3×3 system in two dimensions. We then compute the Laplacian as the trace $\langle \nabla^2 f \rangle_i^{\text{FI}} = u_{i,1} + u_{i,2}$. If we drop the cross-term column from $\mathbf{s}_{ij}$, we recover the block-diagonal, linear-consistent operator, which matches Eq. (13). This is the equivalence we noted earlier. This construction uses only W_{ij} , $\mathbf{B}_i$, and the offsets, completely avoiding kernel derivatives. In practice, we scale the offsets by their root-mean-square length to improve the conditioning of $\mathbf{M}_i^{\text{FI}}$. We verified this formulation against the analytic moment integrals of [8], recovering their uniform-cloud values up to quadrature error. We also confirmed pointwise second-order convergence on disordered clouds.

In this work, we deliberately retain the original linear-consistent LDD operator. We show that when we close the boundary correctly, the boundary closure and not the interior order governs the final accuracy of the solution. The interior operators and the boundary closure we introduce here are complementary and combine easily, as we summarise in Table 2.

3. Boundary closure

3.1. Boundary conditions by type

A simple organising principle underlies the boundary closure. The type of boundary condition, rather than the physical quantity it represents, de-

termines the algebra of its integration into Eq. (13). A Neumann condition prescribes a flux $\nabla p \cdot \mathbf{n} = g$. Because this flux is a known datum independent of the unknown nodal value, we move it to the right-hand side. A Dirichlet or Robin condition prescribes a value, or a value-dependent flux. Since this constraint involves the unknown nodal value itself, it must contribute to the diagonal self-interaction term as well as to the right-hand side. We keep two independent classifications separate. In the first axis, we send fluxes to the right-hand side and values to the diagonal. In the second axis, we either prescribe boundary representations or detect them from the point cloud. We name the two closures derived below after their algebra: the flux closure and the value closure. Physically, solid walls and free surfaces are examples of these closures. In addition, we detect the free surface geometrically rather than prescribing it. The generalised Robin condition $\nabla p \cdot \mathbf{n} + \alpha p = \gamma$ unifies this algebraic view. The flux closure is the limit as $\alpha \rightarrow 0$, whereas the value closure is the limit as $\alpha \rightarrow \infty$. A finite value of α splits the boundary datum between the diagonal and the right-hand side, as we show in Section 5.5. Thus, pairing a wall with a flux and a free surface with a value is specific to the pressure Poisson problem. For the velocity field, this pairing reverses. The no-slip wall becomes a value closure, and the stress-free surface becomes a flux closure, as we discuss in Section 8. We write the unified assembled row as:

$$\sum_j W_{ij} w_{ij} p_{ij} - \sigma_i S_i p_i = b_i, \quad (22)$$

using the weight $w_{ij} = 1 - \mathbf{x}_{ij} \cdot \mathbf{V}_i$. Here, the correction vector $\mathbf{V}_i$ absorbs the Neumann and free-surface geometry, which we derive below. In difference

form, we assemble the sparse matrix and the right-hand side as:

$$A_{ij} = W_{ij} w_{ij} \quad (j \neq i), \quad (23)$$

$$A_{ii} = - \sum_j W_{ij} w_{ij} - \sigma_i S_i, \quad (24)$$

$$b_i = \frac{\nabla^2 p_i}{N_i} + b_{N,i} - \sigma_i S_i p_{\text{target}}. \quad (25)$$

We normalise the source term by N_i as it enters the equations. In contrast, the Neumann flux term $b_{N,i}$ is already normalised, and we do not divide it by N_i . The free-surface term contributes $-\sigma_i S_i$ to the diagonal in Eq. (24) and $-\sigma_i S_i p_{\text{target}}$ to the right-hand side in Eq. (25).

3.2. Flux closure (Neumann) by tangential projection

We present two flux closures for the Neumann boundary. In this section, we detail the tangential projection scheme. It is exact for linear fields but only imposes the flux value. As a consequence, the resulting stencil remains one-sided in the normal direction, which limits its accuracy. This limitation motivates the algebraic-ghost closure that we introduce in Section 3.3.

We do not generate additional nodes on the Neumann boundary. Instead, we activate the modification on any node whose support the boundary truncates, within a distance of order h . Such a node interacts with K nearby boundary segments. Each segment has an outward unit normal $\mathbf{n}_k$, a foot point $\mathbf{x}_{f,k}$, and a prescribed flux g_k . A tangential projector $\mathbf{P}_{\text{tan}}$ removes the boundary-normal directions from the reconstruction. The known flux then supplies the unresolved normal contribution. We introduce two projectors for this purpose.

In the averaged-normal projector, referred to as AN, we build a single effective normal from the boundary normals and project it out:

$$\mathbf{n}_{\text{eff}} = \frac{\sum_k \beta_k \mathbf{n}_k}{\left| \sum_k \beta_k \mathbf{n}_k \right|}, \quad (26)$$

$$\mathbf{P}_{\text{tan}} = \mathbf{I} - \mathbf{n}_{\text{eff}} \otimes \mathbf{n}_{\text{eff}}. \quad (27)$$

Here, the proximity weights decay with the distance to the boundary $\delta_{w,k}$:

$$\beta_k = \frac{\max(0, 1 - \delta_{w,k}/h_w)^2}{\sum_m \max(0, 1 - \delta_{w,m}/h_w)^2}. \quad (28)$$

The AN projector is exact for flat and smoothly curved boundaries. However, non-orthogonal corners introduce an error of $O(\Delta)$. In those regions, a single averaged normal cannot span the full normal subspace. The general Gram projector, referred to as GGP, solves this by removing the entire subspace that the boundary normals span. We collect these normals as the columns of a matrix $\mathbf{N} = [\mathbf{n}_1 | \cdots | \mathbf{n}_K]$. We then define the Gram matrix and the projector as:

$$\mathbf{G} = \mathbf{N}^\top \mathbf{N}, \quad (29)$$

$$\mathbf{P}_{\text{tan}} = \mathbf{I} - \mathbf{N} \mathbf{G}^{-1} \mathbf{N}^\top. \quad (30)$$

The projector is symmetric and idempotent. It annihilates every boundary normal, so that $\mathbf{P}_{\text{tan}} \mathbf{n}_k = \mathbf{0}$. Thus, it is exact for arbitrary corner geometries. We perform regularisation near near-parallel boundaries by setting $\mathbf{G} \rightarrow \mathbf{G} + \epsilon \mathbf{I}$ with $\epsilon \sim 10^{-2}$. For a single segment ($K = 1$) and for orthogonal/near-orthogonal normals, the two projectors coincide. For $K = d$ non-coplanar boundaries, the projector becomes $\mathbf{P}_{\text{tan}} = \mathbf{0}$. In this case, the boundary data entirely determines the fully enclosed node.

A key detail of the renormalisation is that we must perform the projection in physical space, on the raw offset $\mathbf{o}_i$, before renormalising by $\mathbf{B}_i$. We project the offset first and only then renormalise it into the correction vector:

$$\mathbf{o}_{w,i} = \mathbf{P}_{\text{tan}} \mathbf{o}_i, \quad (31)$$

$$\mathbf{V}_i = \mathbf{B}_i \mathbf{o}_{w,i}. \quad (32)$$

If we used the alternative ordering $\mathbf{P}_{\text{tan}}(\mathbf{B}_i \mathbf{o}_i)$, it would remove a component along $\mathbf{n}$ instead of the correct $\mathbf{B}_i \mathbf{n}$ direction. These two directions coincide only if $\mathbf{n}$ is an eigenvector of $\mathbf{B}_i$, which does not happen for real point clouds. Only the project-then-renormalise ordering in Eq. (32) preserves linear consistency at the boundary, as we show next.

Using $\mathbf{V}_i = \mathbf{B}_i \mathbf{o}_{w,i}$, the correction weight becomes $w_{ij} = 1 - \mathbf{x}_{ij} \cdot \mathbf{V}_i$. If we consider a linear field $f = \mathbf{a} \cdot \mathbf{x} + b$, where $f_{ij} = \mathbf{a} \cdot \mathbf{x}_{ij}$, the left-hand side of the assembled row evaluates to:

$$\sum_j W_{ij} f_{ij} w_{ij} = \mathbf{a} \cdot \mathbf{o}_i - \mathbf{a} \cdot \mathbf{M}_i \mathbf{V}_i = \mathbf{a} \cdot \mathbf{o}_i - \mathbf{a} \cdot \mathbf{o}_{w,i} = \mathbf{a} \cdot (\mathbf{I} - \mathbf{P}_{\text{tan}}) \mathbf{o}_i, \quad (33)$$

where we use the identity $\mathbf{M}_i \mathbf{B}_i = \mathbf{I}$. This residual represents the part of $\mathbf{o}_i$ that lies in the boundary-normal subspace. The closure cancels this term by placing the prescribed flux on the right-hand side:

$$b_{N,i} = \begin{cases} g_{\text{eff}} (\mathbf{o}_i \cdot \mathbf{n}_{\text{eff}}), & \text{AN, } g_{\text{eff}} = \sum_k \beta_k g_k, \\ (\mathbf{N}^\top \mathbf{o}_i)^\top \mathbf{G}^{-1} \mathbf{g}, & \text{GGP.} \end{cases} \quad (34)$$

We verify this cancellation by writing the Gram projector complement as $\mathbf{I} - \mathbf{P}_{\text{tan}} = \mathbf{N} \mathbf{G}^{-1} \mathbf{N}^\top$. For a linear field, the vector of nodal fluxes is $\mathbf{g} =$

$[\mathbf{n}_k \cdot \mathbf{a}]_k = \mathbf{N}^\top \mathbf{a}$. The residual in Eq. (33) then becomes:

$$\mathbf{a} \cdot (\mathbf{I} - \mathbf{P}_{\text{tan}}) \mathbf{o}_i = \mathbf{a}^\top \mathbf{N} \mathbf{G}^{-1} \mathbf{N}^\top \mathbf{o}_i = (\mathbf{N}^\top \mathbf{o}_i)^\top \mathbf{G}^{-1} \mathbf{g} = b_{\text{N},i}. \quad (35)$$

Thus, the equation $\sum_j W_{ij} f_{ij} w_{ij} = b_{\text{N},i}$ holds for any linear field on any truncated support, including non-orthogonal corners. If we had used the reversed ordering $\mathbf{P}_{\text{tan}}(\mathbf{B}_i \mathbf{o}_i)$, the term $\mathbf{M}_i \mathbf{P}_{\text{tan}} \mathbf{B}_i \mathbf{o}_i \neq \mathbf{o}_{w,i}$ would replace $\mathbf{M}_i \mathbf{V}_i = \mathbf{o}_{w,i}$ in Eq. (33), breaking this exact cancellation. We refer to this method as the projection closure.

The projection closure is correct and exact for linear fields. However, the closure only imposes the first-derivative flux at the boundary. For nonlinear solutions, the normal second derivative $\partial^2 p / \partial n^2$ retains a one-sided support. This leads to poor resolution. As we show in Section 5, the Neumann boundary becomes the accuracy-limiter. According to the boundary-layer mechanism of LeVeque [53, §2.12], an $O(h)$ boundary-row truncation error sits at a node whose column of the inverse operator is $O(1)$. The operator therefore does not damp this error, which pollutes the global solution with a large constant offset. To remedy this, we complete the stencil in the next section.

3.3. Flux closure (Neumann) by algebraic ghosting

Stencil completion removes the limitation of the projection closure. Instead of projecting out the unresolved normal direction, we fill the support with image samples. This turns the normal second derivative $\partial^2 p / \partial n^2$ into a genuine two-sided difference, with the prescribed flux setting the values of the image points. This is the basic ghost-node, or image-particle, construction [9, 53]. In this work, we package this construction for the renormalised

operator without adding new unknowns. Consider a Neumann node i with nearby boundary segments $\{(\mathbf{n}_k, \mathbf{x}_{f,k}, g_k)\}$. Let R_k denote reflection across the plane passing through $\mathbf{x}_{f,k}$ with normal $\mathbf{n}_k$. We define the reflection group as the set of all non-empty compositions of these reflections R_k . For a single segment ($K = 1$), the group is simply $\{R_1\}$. For an orthogonal corner ($K = 2$), the group is $\{R_1, R_2, R_2 \circ R_1\}$, where the last composition represents the double reflection that fills the far quadrant. For a non-orthogonal wedge, the two planes generate an infinite group. In this case, we truncate the group to the single reflections and the first-order double reflection $R_2 \circ R_1$. Although this truncation does not reproduce the exact image set of the wedge, it is empirically sufficient. This restores the symmetry of the moment tensor $\mathbf{M}_i$ and recovers the convergence rates reported in Section 5.7. We reflect every neighbour j as well as the node itself. Under a composition, a source point $\mathbf{x}_s$ maps to $\mathbf{x}_s \mapsto \mathbf{x}_s - 2\sigma_k \mathbf{n}_k$, where we define σ_k as $(\mathbf{x}_s - \mathbf{x}_{f,k}) \cdot \mathbf{n}_k$. We assign a Neumann-consistent value to the corresponding image point: $p_{\text{ghost}} = p_s + \sum_k (-2\sigma_k g_k)$. We append every image point within the support radius h to the stencil with its kernel weight. These points enter the calculations of $\mathbf{M}_i$, $\mathbf{o}_i$, and N_i . This process restores a near-symmetric support. We can normalise the completed row using either of the forms from Section 2. We prefer the sum normalisation in Eq. (14) because it restores close to second-order convergence at straight Neumann boundaries. At non-orthogonal corners, we retain the smooth-naive form, as shown in Section 5.3. The Laplacian row couples the image point back to its source node p_s (which is either a neighbour or node i itself) using the standard weight, and we move the constant flux increment to the right-hand side. We verified that

two specific details are necessary. First, the self-reflection of node i provides the closest and highest-weight image point. The diagonal contribution cancels out, while the flux increment remains on the right-hand side. Second, the double reflection at corners fills the otherwise empty quadrant. We refer to this scheme as the algebraic-ghost closure. The scheme preserves linear exactness. For a linear field $f = \mathbf{a} \cdot \mathbf{x} + b$, the flux is $g = \mathbf{a} \cdot \mathbf{n}$. The assigned image value reproduces the true field at the mirrored point:

$$p_{\text{ghost}} = f(\mathbf{x}_s) - 2\sigma(\mathbf{a} \cdot \mathbf{n}) = \mathbf{a} \cdot (\mathbf{x}_s - 2\sigma\mathbf{n}) + b = f(\mathbf{x}_s - 2\sigma\mathbf{n}) = f(\mathbf{x}_{\text{ghost}}), \quad (36)$$

Thus, the completed support acts as an ordinary, linear-exact renormalised stencil. Unlike the projection closure, the two-sided support resolves the normal curvature. The boundary is no longer the bottleneck. This reflection plane approximation is exact for straight boundary segments. For curved boundaries, we use a finely faceted polygon, leaving a local curved-plane generalisation for future work. We apply this closure only to Neumann nodes. The free surface, the contact lines, and the interior retain the single code path of Sections 3.2 and 3.4.

These ghost nodes are algebraic and ephemeral. We form them on the fly within the local neighbour sum of a single row. They contribute to that row's moment tensor and coefficients before we discard them. We do not store ghost particles, allocate global memory beyond row-local arrays, or extend and re-search the global neighbour list. This provides a key advantage over dummy-particle treatments, in which layers of physical boundary particles are allocated and maintained, significantly increasing the cost of neighbour searches.

3.4. Value closure (Dirichlet/Robin) and free-surface detection

Following the main principle of this work, the value closure integrates the Dirichlet or Robin condition directly into the diagonal. The primary application is the free surface. We detect this value boundary directly from the cloud rather than prescribing it beforehand. At the free surface, the field satisfies $p = p_{\text{target}}$. This target value is zero for a clean surface and $\gamma\kappa$ when surface tension is present. We do not place surface particles. Instead, every node with an air-side truncation receives this boundary condition. We convert the Dirichlet condition into a Robin form by a Taylor expansion to the surface at a distance δ along the outward normal $\mathbf{n}_s$:

$$\nabla p_i \cdot \mathbf{n}_s = \frac{p_{\text{target}} - p_i}{\delta}. \quad (37)$$

This formulation produces the diagonal modification in Eq. (22) instead of a pure right-hand-side source. We obtain the surface geometry from the point cloud by identifying the support deficiency that the Neumann boundary cannot explain. Specifically, we compute it from the projected offset $\mathbf{o}_{w,i}$. This ensures that corner geometries do not contaminate the surface normal. We calculate the residual vector, the outward normal, and the surface distance as:

$$\mathbf{r}_i = \mathbf{B}_i \mathbf{o}_{w,i}, \quad (38)$$

$$\mathbf{n}_s = -\mathbf{r}_i/|\mathbf{r}_i|, \quad (39)$$

$$\delta = |\mathbf{o}_{w,i} \cdot \mathbf{n}_s|/S_i. \quad (40)$$

With these natural estimates, the Robin enforcement strength $-(\mathbf{o}_{w,i} \cdot \mathbf{n}_s)/\delta$ simplifies to a purely geometric quantity. If we write $\mathbf{n}_s = -\mathbf{r}_i/|\mathbf{r}_i|$ and

$\mathbf{r}_i = \mathbf{B}_i \mathbf{o}_{w,i}$, the numerator is:

$$\mathbf{o}_{w,i} \cdot \mathbf{n}_s = -\frac{\mathbf{o}_{w,i} \cdot \mathbf{r}_i}{|\mathbf{r}_i|} = -\frac{\mathbf{o}_{w,i} \cdot \mathbf{B}_i \mathbf{o}_{w,i}}{|\mathbf{r}_i|} < 0. \quad (41)$$

The positive-definiteness of $\mathbf{B}_i$ guarantees a negative sign. This reflects the fact that the offset points inward while the normal points outward. The distance estimate is $\delta = |\mathbf{o}_{w,i} \cdot \mathbf{n}_s|/S_i = (\mathbf{o}_{w,i} \cdot \mathbf{B}_i \mathbf{o}_{w,i})/(|\mathbf{r}_i| S_i)$. Upon division, the common factor $(\mathbf{o}_{w,i} \cdot \mathbf{B}_i \mathbf{o}_{w,i})/|\mathbf{r}_i|$ cancels out completely between the numerator and denominator:

$$-\frac{\mathbf{o}_{w,i} \cdot \mathbf{n}_s}{\delta} = \frac{(\mathbf{o}_{w,i} \cdot \mathbf{B}_i \mathbf{o}_{w,i})/|\mathbf{r}_i|}{(\mathbf{o}_{w,i} \cdot \mathbf{B}_i \mathbf{o}_{w,i})/(|\mathbf{r}_i| S_i)} = S_i. \quad (42)$$

Thus, the enforcement strength is exactly the total kernel weight S_i . This quantity is never singular, scales automatically, and is free of any penalty parameters. We detect the free surface without using a threshold by defining a dimensionless indicator:

$$\lambda_i = |\mathbf{r}_i| \Delta. \quad (43)$$

This indicator is close to zero in the interior of the domain. At the surface, it becomes $O(1)$. We map it to the range $[0, 1]$ using a smooth activation function $\sigma_i = \sigma(\lambda_i)$. We recommend a cubic smoothstep function:

$$\sigma_i = S\left(3\left(\lambda_i - \frac{2}{3}\right)\right), \quad (44)$$

where we define the smooth step function as $S(t) = \begin{cases} 0, & t \leq 0, \\ 3t^2 - 2t^3, & 0 < t < 1, \\ 1, & t \geq 1, \end{cases}$

which remains zero for $\lambda_i \leq \frac{2}{3}$ to leave interior nodes untouched. It reaches unity for $\lambda_i \geq 1$, and introduces no free parameters. We then remove the

normal component in a regularised manner that avoids dividing by zero, and renormalise the result into the correction vector and weight:

$$\mathbf{o}_{*,i} = \mathbf{o}_{w,i} - \sigma_i \frac{(\mathbf{o}_{w,i} \cdot \mathbf{r}_i) \mathbf{r}_i}{|\mathbf{r}_i|^2 + \eta^2}, \quad (45)$$

$$\mathbf{V}_i = \mathbf{B}_i \mathbf{o}_{*,i}, \quad (46)$$

$$w_{ij} = 1 - \mathbf{x}_{ij} \cdot \mathbf{V}_i. \quad (47)$$

The diagonal of the row receives a contribution of $-\sigma_i S_i$, while the right-hand side receives the term $-\sigma_i S_i p_{\text{target}}$. Here, we evaluate p_{target} at the detected surface point $\mathbf{x}_i + \delta \mathbf{n}_s$. In this way, we integrate the free-surface condition as a value-dependent Robin flux. Its known part enters the right-hand side, and its value-dependent part goes to the diagonal of the same collocation row. This is exactly the treatment used for the Neumann flux. We thus recast every boundary condition as a generalised flux and fold it directly into the operator. Because of this unified treatment, we name the method boundary-integrated or flux-integrated.

3.5. Contact lines and the unified equation

A node at a contact line acts as both a Neumann node and a surface node. Because we compute the surface residual from the projected offset $\mathbf{o}_{w,i}$, it lies in the boundary-tangential subspace. As a result, the two projections do not interfere with one another. This achieves a clean composition between the Neumann projection of Section 3.2 and the surface removal of Section 3.4. We produce a single vector $\mathbf{o}_{*,i}$ and a single correction vector $\mathbf{V}_i$. We then assemble Eq. (22) with the Neumann flux on the right-hand side and the surface value term on the diagonal. Interior nodes have $\sigma_i = 0$ and no

boundary interactions, so they simplify to Eq. (13). Neumann-only nodes reduce to the equations in Section 3.2, and surface-only nodes reduce to those in Section 3.4. All classes of nodes thus retain the single-code-path property. Note that a pure-Neumann domain is singular up to an archetype scalar constant. We must close such systems with a zero-mean constraint instead of pinning a single point. However, any free-surface node with $\sigma_i > 0$ automatically removes this null space.

3.6. A remark on consistency and convergence

The closures inherit the linear consistency of the interior operator. For a linear field $f = \mathbf{a} \cdot \mathbf{x} + b$, the gradient in Eq. (5), the Laplacian in Eq. (13), the projection closure in Eq. (34), and the algebraic-ghost closure all reproduce the exact solution. This holds for truncated supports and corner rows alike, as we verify to round-off error in Section 4. Furthermore, the free-surface cancellation in Eq. (42) is an exact algebraic identity. However, as noted in Section 2, linear consistency does not guarantee second-order pointwise consistency under fixed-relative disorder. In practice, the boundary closure governs the actual convergence rate. The value closure supraconverges. On the other hand, the flux-only Neumann closure leaves the normal second derivative one-sided. The boundary thus becomes the primary accuracy bottleneck due to the undamped boundary column mechanism described by LeVeque [53, §2.12]. The algebraic-ghost closure resolves this by supplying the missing normal curvature. We quantify these observations in the following sections. For the formal consistency and convergence proofs of the underlying operators, we refer readers to [1, 47].

4. Operator approximation results

We verified the discrete operators against analytic ground truths on both regular and scattered point clouds. To measure particle disorder, we use the chaos level c defined in [1]. We displace each node by a random vector with components in the interval $[-c\Delta/2, c\Delta/2]$. We report scattered results as the median over at least twenty random realisations. We choose the normalised root-mean-square error, $\text{Error} = \|\langle f \rangle - f\|_2 / \|f\|_2$, as our error metric, following [1].

The renormalised gradient $\langle \nabla f \rangle_i$ in Eq. (5) and the Laplacian in Eq. (13) are exact for linear fields on all tested clouds. This accuracy extends to truncated boundary rows and the non-orthogonal wedge, matching analytical values to within $\sim 10^{-15}$ and $\sim 10^{-13}$ respectively. The Gram-projector algebra itself is exact to within $\sim 10^{-16}$. The projector $\mathbf{P}_{\text{tan}}$ is symmetric and idempotent, satisfying $\mathbf{P}_{\text{tan}} \mathbf{n}_k = \mathbf{0}$ for every boundary normal. We find that AN and GGP coincide when $K = 1$ or when the normals are orthogonal. With three non-coplanar boundaries, the projector becomes $\mathbf{P}_{\text{tan}} = \mathbf{0}$. The Neumann linear consistency condition, $\sum_j W_{ij} w_{ij} f_{ij} = b_{N,i}$, holds to within $\sim 10^{-17}$. Our tests confirm that the project-then-renormalise ordering in Eq. (32) is essential. Reversing this order breaks consistency, introducing errors of $O(10^{-3})$ on a jittered cloud. The free-surface cancellation in Eq. (42) holds up to machine precision. As a result, we recover the hydrostatic linear Poisson problem on a wedge tank to within $\sim 10^{-14}$ using GGP boundaries. Finally, the algebraic-ghost closure preserves linear exactness to within $\sim 10^{-15}$.

We confirmed the $2/d$ factor of the smooth-naive normalisation in Eq. (9)

using a three-dimensional constant-Laplacian test. Without this factor, the bare $\text{tr } \mathbf{B}_i$ normalisation is off by a factor of 1.5. The sum operator uses the denominator normalisation N_i^{den} from Eq. (14). Because it is exact for isotropic quadratics, it is about 8 to 15% more accurate at typical resolutions. The advantage becomes more pronounced under anisotropy. As shown in Fig. 16, the sum-to-smooth-naive error ratio decreases monotonically from 1.00 on a regular grid to 0.73 at 80% chaos. We offer this as a refinement. We retain the smooth-naive operator as the default choice because of its lower computational cost.

The pointwise approximation of the operator was evaluated using the Franke test function [54] on a unit square. This follows the methodology of [1]. In Fig. 1, we plot the normalised error of the renormalised Laplacian against the spacing Δ . We compare a regular lattice with disordered clouds at 30, 60, and 90% chaos. We also include two second-order baselines: the GFDM operator and the Full Inverse operator of Asai et al. [8]. On a regular grid, all these operators achieve second-order accuracy. Under particle disorder, the renormalised operator is linear-consistent. However, the pointwise convergence order drops as chaos increases. It goes from 2.0 on the lattice to about 0.7, 0.2, and 0.1 at 30, 60, and 90% chaos, respectively. This matches the $O(1)$ truncation error noted in Section 2. In contrast, the GFDM operator maintains the second-order rate. The Full Inverse operator tracks the GFDM operator closely, matching it to within plotting accuracy. This is because the Full Inverse operator restores the cross-derivative coupling to the renormalised stencil. The comparison shows that the loss of consistency is due to the omitted cross terms rather than the renormalised framework

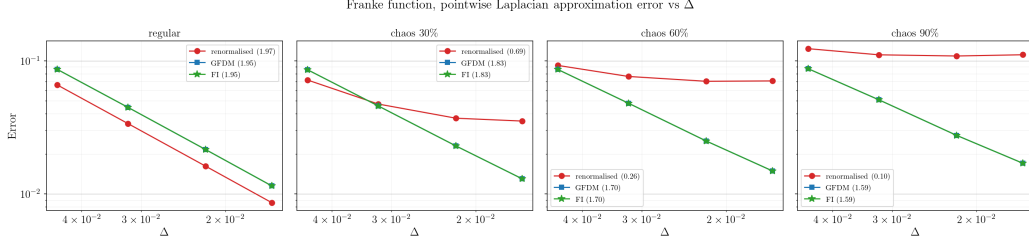


Figure 1: Pointwise Laplacian approximation error for the Franke function against spacing Δ on regular and disordered point clouds.

itself. To recover second-order accuracy, we can reinstate the cross terms at the expense of a small local solve per node. Alternatively, we can increase the support radius to recover the convergence order of the renormalised operator. At 60% chaos, the order rises to about 1.0 when we grow the support from 2.5Δ to 3.5Δ . It reaches about 1.8 at 5Δ , though this requires a wider stencil.

The support is chosen to remain small because the global solution to the boundary value problem is still supraconverged, as demonstrated in Section 5.

5. Boundary value problem results

The closures were tested on several Poisson benchmarks in irregular domains. These cases extend the work of [9, 13] and [1]. The discretisation is refined by halving Δ at each step. Scattered point clouds are tested at chaos levels of 30, 60, and 90%, with results averaged over at least twenty realisations using the median. The second-order GFDM operator from Section 2.2 serves as the main baseline. Its Neumann condition is imposed using the exact constraint row, with the penalty form kept for comparison. For the interior operator comparisons, the Full Inverse operator described in Sec-

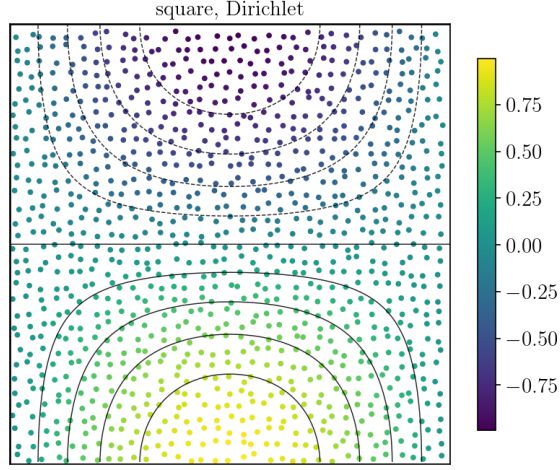


Figure 2: Qualitative solution map for the square domain with a Dirichlet condition and the value closure on a flat boundary.

tion 2.3 is also included. The classical SPH Laplacian fails to converge on irregular clouds, as established in [1]; those tests are not repeated here. In Figs. 2–5, performance on each domain is previewed. The numerical solution on the scattered cloud is plotted against the contours of the exact solution. The square with a Dirichlet condition, the star with a Dirichlet condition, the flower with a Robin condition, and the wedge tank with Neumann boundaries and a detected free surface are shown.

5.1. Square domain with a Dirichlet condition

The first baseline is a unit square with a manufactured trigonometric solution and the Franke test function [54]. In Fig. 6, the renormalised closure is compared against the GFDM baseline and the Full Inverse operator [8] at three chaos levels. Pointwise consistency is retained by the two second-order operators. The method maintains a convergence rate near 1.77 across

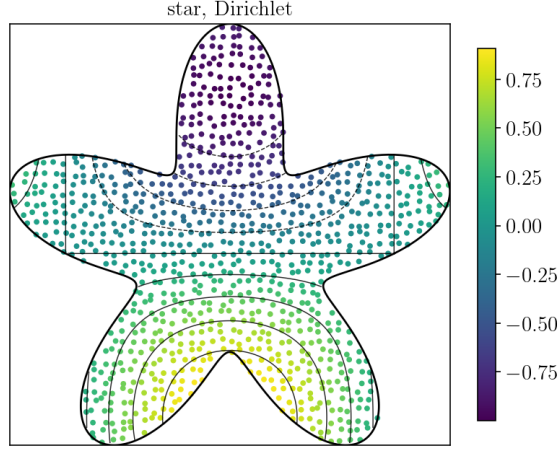


Figure 3: Qualitative solution map for the star domain with a Dirichlet condition and the curved value closure.

all disorder levels, with the curves coinciding. In contrast, the renormalised operator achieves supraconvergence at the linear-consistent limit. It achieves a convergence rate between 1.4 and 1.5 that degrades only slightly with increasing disorder. The primary advantage of the proposed method is not its interior convergence rate. Indeed, a fully second-order operator attains higher accuracy under strong disorder. Rather, the value lies in the boundary closure, which we examine in the next benchmarks, as well as in the robustness and the absence of tuning parameters. The second-order interior operator and the boundary closure introduced here are complementary, and we can combine them easily.

5.2. Neumann boundaries by projection, algebraic ghost and GFDM constraint row

We use an all-Neumann unit box closed by a zero-mean constraint to isolate the Neumann treatment. In Table 1 and Fig. 7, we compare four

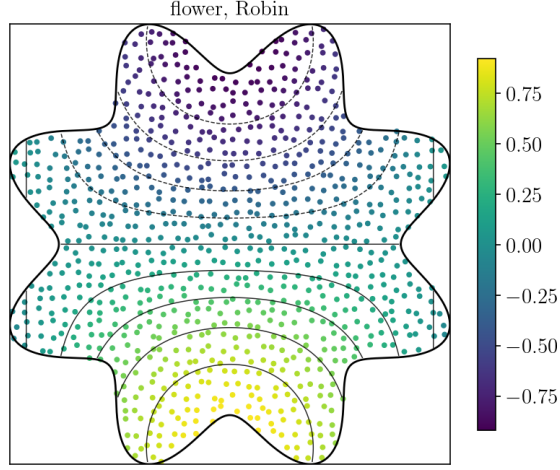


Figure 4: Qualitative solution map for the flower domain with a Robin condition.

distinct closures: the FIRM projection closure, the FIRM algebraic-ghost closure, the GFDM exact constraint row, and the GFDM penalty form. The two flux-only closures (FIRM projection and the GFDM constraint row) exhibit similar behaviour. Both impose the flux exactly, but leave the normal second derivative one-sided. Consequently, we obtain slow convergence, with an order of about 0.9. They also carry a large error constant, with the Neumann boundary dominating the error. The penalty form is virtually indistinguishable from the constraint row. This similarity confirms that the issue is not a soft flux implementation. The algebraic-ghost closure completes the stencil. As a result, we achieve an error reduction of about eight times at the finest resolution, yielding an error of 4.7×10^{-3} compared to 3.9×10^{-2} . Because the all-Neumann box is singular up to an archetype scalar constant, how we handle the null space heavily influences the convergence rate for each closure. This makes the box a poor case for measuring convergence rates.

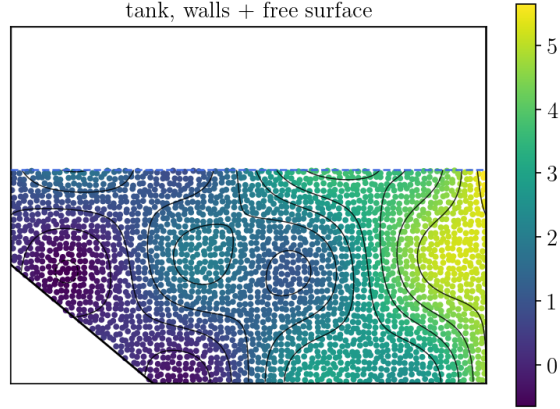


Figure 5: Qualitative solution map for the wedge tank with GGP Neumann boundaries, the algebraic-ghost closure, and a geometry-detected free surface.

Therefore, we report the convergence rate on the well-posed wedge tank in Section 5.7. On that domain, the ghost closure lifts the Neumann-region order from about 1.0 to between 1.3 and 1.8. Thus, the Neumann boundary converges at roughly the same rate as the interior. Importantly, the algebraic-ghost closure outperforms the GFDM constraint row because completing the stencil supplies the local curvature. An exactly imposed flux alone does not provide this information. A fully high-order Neumann treatment would require the staggered KKT approach of Trask et al. [29], which we leave for future research.

5.3. Normalisation of the ghost closure

How we normalise the completed row determines the convergence order of the algebraic-ghost closure. We isolate this effect on a square domain with a single straight Neumann boundary at $y = 0$ and Dirichlet conditions on the other three edges. This problem is well-posed and free of the corner

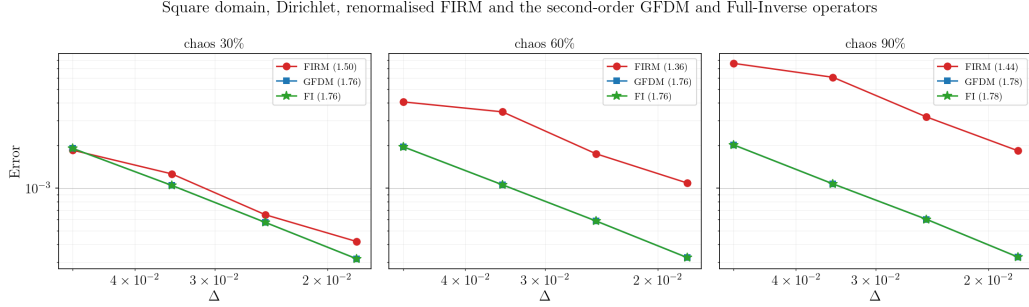


Figure 6: Convergence of the Dirichlet Poisson solution error on a square domain under different chaos levels.

Table 1: Poisson solution error on the all-Neumann box under 30% chaos for different boundary closures.

Δ	FIRM projection	FIRM ghost	GFDM constraint	GFDM penalty
0.050	1.05×10^{-1}	1.1×10^{-2}	1.22×10^{-1}	1.22×10^{-1}
0.035	7.2×10^{-2}	1.6×10^{-2}	8.8×10^{-2}	8.8×10^{-2}
0.025	6.0×10^{-2}	2.2×10^{-3}	6.7×10^{-2}	6.7×10^{-2}
0.018	3.9×10^{-2}	4.7×10^{-3}	4.7×10^{-2}	4.7×10^{-2}

truncation discussed in Section 3.3. In Fig. 8, we report the error along the Neumann edge against the spacing Δ . The projection closure achieves convergence at an order of about 0.8. Using the ghost closure with the smooth-naive normalisation from Eq. (9) improves the error constant by an order of magnitude. However, the error eventually reaches a plateau at about 1.4. The ghost closure with the sum normalisation from Eq. (14) is exact for $|\mathbf{x}|^2$. This removes the plateau, and convergence continues at a rate of about 1.8, approaching second-order accuracy. On the completed and near-symmetric support, the two formulations differ only due to residual anisotropy. By re-

Neumann boundaries, mirror-ghost and flux-only closures

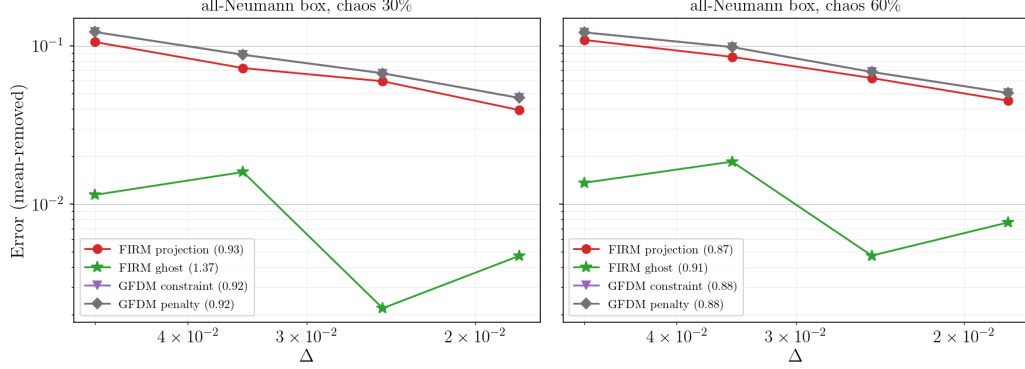


Figure 7: Comparison of Poisson solution error under Neumann boundary conditions on a square box for different closures.

producing the curvature of $|\mathbf{x}|^2$ exactly, the sum normalisation supplies the normal second derivative. In contrast, the smooth-naive normalisation leaves this derivative biased. Therefore, we prefer the sum normalisation for the ghost closure on straight and smoothly curved Neumann boundaries. At a non-orthogonal corner, the truncated reflection group of Section 3.3 caps the convergence rate. In those regions, we retain the more robust smooth-naive formulation, as shown in Section 5.7.

5.4. Star domain with Dirichlet and Neumann conditions

We test the closures on a curved boundary using a star-shaped domain $r(\theta) = 0.5 + 0.2 \sin 5\theta$, following [9], with a trigonometric manufactured solution. We represent the boundary with fine facets, keeping the residual faceting error well below the solution error. For the Dirichlet boundary condition, we apply the value closure from Section 3.4 using the local boundary normal and distance. This configuration achieves supraconvergence at an

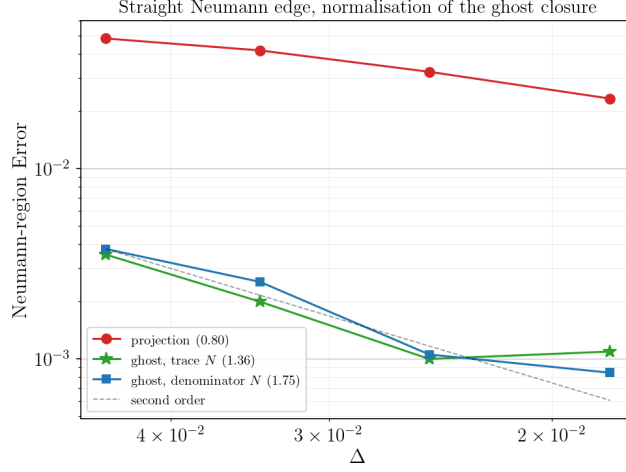


Figure 8: Neumann boundary error convergence along a straight edge under the projection and ghost closures.

order of about 1.85, outperforming the GFDM baseline which converges at about 1.4. For the Neumann condition, the projection closure acts as the main bottleneck, converging at about 0.8. The algebraic-ghost closure improves both the error constant and the convergence order, lifting it to about 1.7. The GFDM constraint row closely tracks the projection closure at about 0.9, matching the findings on the box domain. We collect these convergence curves in Fig. 9.

5.5. Flower domain with a Robin condition

We use a flower-shaped domain to test the value-dependent flux closure with a Robin condition $\partial u / \partial n + \alpha u = f$. Because the boundary value is unknown, we eliminate it using a local Taylor expansion along the normal direction to the boundary: $u(\mathbf{x}_f) = u_i + \delta \partial u / \partial n$. We substitute this relation into the Robin condition, $\partial u / \partial n + \alpha u(\mathbf{x}_f) = f$, and solve for the normal

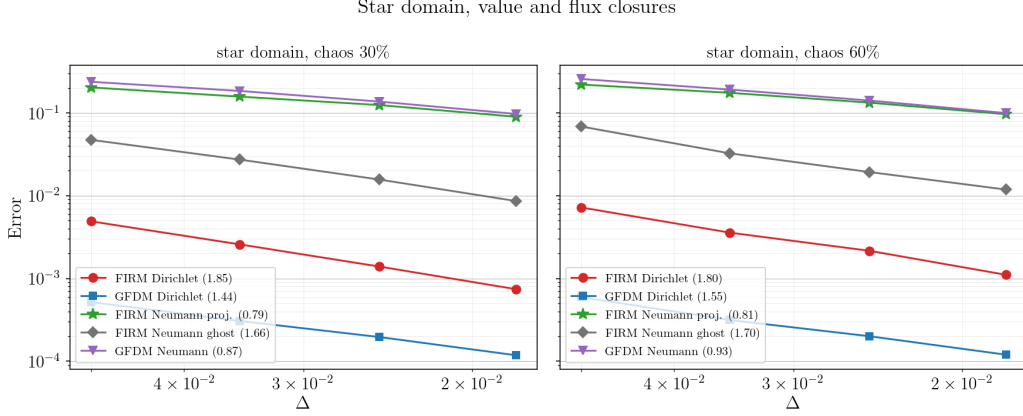


Figure 9: Convergence of Dirichlet and Neumann Poisson solutions on a curved star domain under boundary disorder.

flux:

$$g = \frac{\partial u}{\partial n} = \frac{f - \alpha u_i}{1 + \alpha \delta}. \quad (48)$$

The known term, $f/(1 + \alpha \delta)$, enters the right-hand side of the flux equations, matching Eq. (34). The value-dependent term, $-\alpha u_i/(1 + \alpha \delta)$, is linear in the unknown value u_i . We move it to the diagonal of the row, where $\alpha/(1 + \alpha \delta)$ times the geometric weight $(\mathbf{N}^\top \mathbf{o}_i)^\top \mathbf{G}^{-1} \mathbf{1}$ makes a contribution. Equation (48) simplifies to the pure Neumann closure in the limit as $\alpha \rightarrow 0$. In that limit, we obtain $g \rightarrow f$, with no diagonal modification. As $\alpha \rightarrow \infty$, we recover the Dirichlet value closure, where $g \rightarrow (f/\alpha - u_i)/\delta$. This formulation remains exact for linear fields on straight boundary segments. The realised convergence rate ranges from about 0.9 at 30% chaos to 1.1 at 90% chaos. This is consistent with a flux-type boundary treatment, as Figure 10 shows.

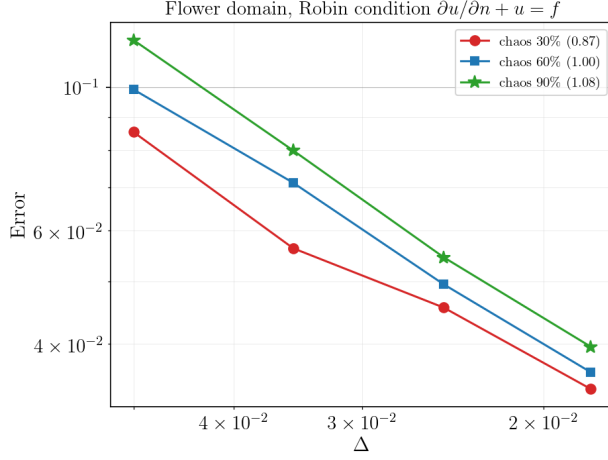


Figure 10: Robin boundary condition convergence on a curved flower domain against the spacing Δ for different chaos levels.

5.6. Free-surface detection from the cloud

In this benchmark, we treat the upper boundary of a polygonal tank as an unknown free surface. We detect it directly from the point cloud using the indicator λ_i and the smoothstep activation function described in Section 3.4. We do not use surface particles or prescribed boundary geometry. The detector achieves a clear separation between the interior and surface nodes. For example, the interior indicator value is $\lambda^{(90)} \approx 0.2$, which sits well below the activation threshold of $\frac{2}{3}$. The outermost surface ring yields values between $\lambda \approx 0.7$ and 0.9 . The interior activation is zero. We observed no false activations at any of the tested resolutions or disorder levels. Furthermore, the recovered surface normal matches the analytical vertical direction to within a few percent. The geometry-detected closure closely tracks the exact-boundary reference. The error in the surface region for the detected closure converges at an order of about 1.5. This is close to, but

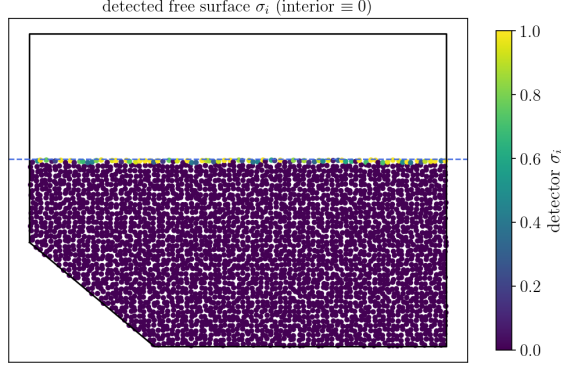


Figure 11: Geometry-detected activation σ_i on the jittered tank, which is zero in the interior and rises to unity at the surface.

slightly below, the exact reference rate of 2.2, and requires no tuning parameters. We plot the detected activation field and compare convergence rates in Figs. 11 and 12. For comparison, we also tested a rational activation function, $\sigma = \lambda^2/(\lambda^2 + c^2)$, with a fixed parameter c . This alternative function false-triggers in the interior under strong particle disorder. The smoothstep function remains pinned at zero, which justifies this parameter-free choice.

5.7. Non-orthogonal wedge with a free surface

We use a polygonal tank with a non-orthogonal slant-and-floor wedge and a free surface to test all closures together on a complex manufactured field: $p^* = \sin \pi x \cos \pi y + 0.3(x^2 + y^2)$. At the wedge boundary, the GGP projector is exact for linear fields, yielding a relative error of $\sim 10^{-14}$. In contrast, the AN projector introduces an $O(\Delta)$ leakage, with a relative error of $\sim 3 \times 10^{-2}$. This matches the prediction in Section 3.2. For the manufactured field, the error concentrates near the solid boundaries and the corner wedge. It vanishes near the free surface. The Neumann boundary thus acts as the

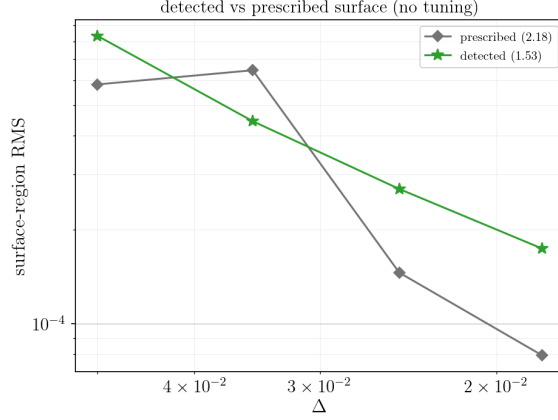


Figure 12: Convergence of the detected surface closure compared to the prescribed reference without using any tuning parameters.

main convergence bottleneck under the projection closure, where we obtain convergence at an order of 1.0. Replacing the projection closure with the algebraic-ghost closure reduces the error in the Neumann region by about an order of magnitude. At the finest resolution, the error drops from 1.0×10^{-2} to 7.6×10^{-4} . Furthermore, the ghost closure lifts the convergence rate in the Neumann region to about 1.3 over the full range and about 1.8 in the asymptotic range. The Neumann region now converges at roughly the same rate as the interior and the Dirichlet boundary, which converge at rates of 1.4 to 1.8. Thus, the Neumann boundary is no longer the bottleneck. We average these errors over six random seed realisations. We find that both the corner double reflection and the self-reflection of boundary nodes are necessary to achieve this rate. We present these convergence results in Fig. 13, and show the spatial distribution of the error in Fig. 14.

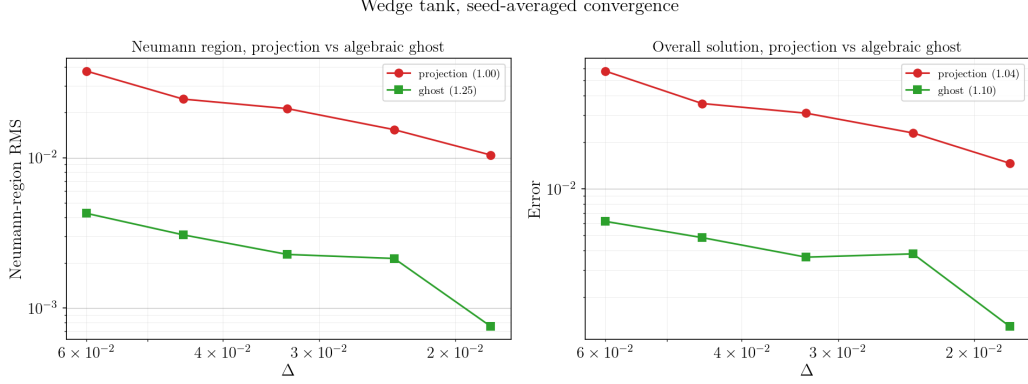


Figure 13: Neumann-region and overall Poisson solution errors on the wedge tank for different closures.

5.8. Kernel, support radius and disorder robustness

The operators are kernel-agnostic. Linear exactness holds for the poly6 $(1 - q^2)^3$, the spiky $(1 - q)^3$, and the Wendland-C2 kernels at any support radius. However, this independence does not extend to the solution accuracy. At the same support radius, the poly6 kernel weights neighbours most evenly, making it the most accurate choice at small support sizes. For instance, at $h = 2.0\Delta$, the Poisson solution error is 2.3×10^{-3} using the poly6 kernel compared to 6.6×10^{-3} with the spiky kernel. The spiky kernel only becomes competitive when the support is relatively large, around $h \geq 2.5\Delta$. This, however, removes the advantage of a small support. We find the default setting of $h = 2.5\Delta$ to be highly robust. It provides about fifteen interior neighbours and at least four boundary neighbours at 60% chaos. We find that a smaller support of $h \in [2.0, 2.25]\Delta$ works well for quasi-uniform point clouds, as the algebraic-ghost closure relieves the boundary truncation constraint. Reducing the support below $h \leq 2.2\Delta$ runs the risk of falling below

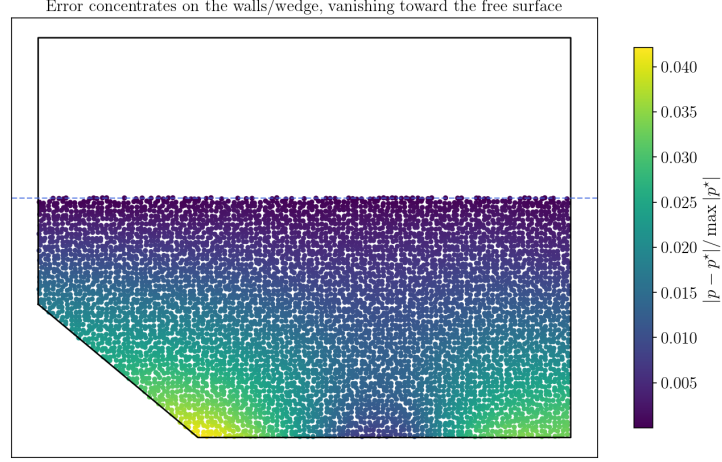


Figure 14: Pointwise Poisson solution error on the wedge tank under GGP wall and prescribed surface closures.

the two-dimensional rank limit of three neighbours under strong disorder, which leads to numerical noise. Across chaos levels from 0 to 80%, the operators remain exact for linear fields. The condition number of the moment matrix $\mathbf{M}_i$ rises smoothly from 1.9 to about 6, and $\mathbf{M}_i$ remains invertible. The smoothstep activation function never false-triggers in the interior. Additionally, the benefit of the denominator normalisation grows with anisotropy. We collect these trends in Figs. 16 and ??.

6. Application to the Helmholtz–Hodge decomposition

The discrete Helmholtz-Hodge decomposition provides a natural application of the boundary closure. Here, we split a vector field into a divergence-free part and the gradient of a scalar potential. The pressure projection in incompressible flow solvers is the best-known special case. However, this decomposition is general. It arises wherever we must project a field onto its

Kernel shape & support radius robustness (chaos 60%)

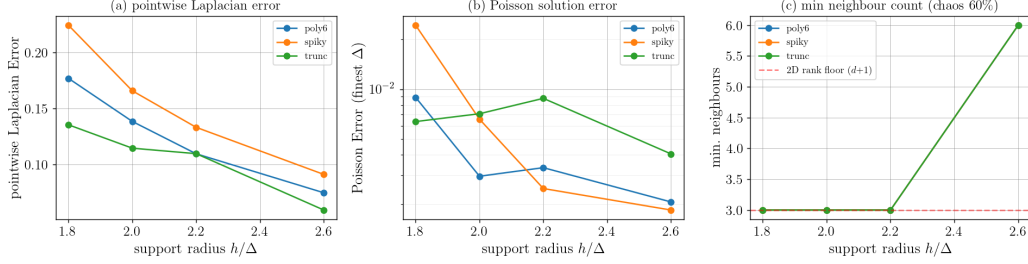


Figure 15: Effect of kernel shape and support radius on the pointwise Laplacian error, Poisson solution error, and minimum neighbour count.

solenoidal component. We keep the discussion in these general terms. If we consider a vector field $\mathbf{u}^*$, we compute the decomposition $\mathbf{u}^* = \mathbf{u} + \nabla\phi$ with $\nabla \cdot \mathbf{u} = 0$ by solving a scalar Poisson problem for the potential ϕ :

$$\mathbf{L} \phi = \langle \nabla \cdot \mathbf{u}^* \rangle, \quad (49)$$

where we define the projected velocity as $\mathbf{u} = \mathbf{u}^* - \langle \nabla\phi \rangle$. In this equation, $\mathbf{L}$ represents the renormalised Laplacian with the boundary closure of Section 3. The divergence term $\langle \nabla \cdot \mathbf{u}^* \rangle_i = \sum_c \mathbf{e}_c \cdot \langle \nabla u_c^* \rangle_i$ is the trace of the component-wise renormalised gradient in Eq. (5). The potential satisfies a Neumann condition at solid walls, and a value condition at the free surface, following the boundary-condition-by-type principle of Section 3. A key concern for meshless, collocated discretisations is that the operators are not mutually adjoint. Specifically, the single-sum Laplacian does not exactly compose the discrete divergence and gradient operators: $\mathbf{L} \neq \mathbf{D}\mathbf{G}$, where $\mathbf{D}$ and $\mathbf{G}$ represent the divergence and gradient matrices. Consequently, subtracting the gradient does not completely annihilate the divergence. We evaluate the

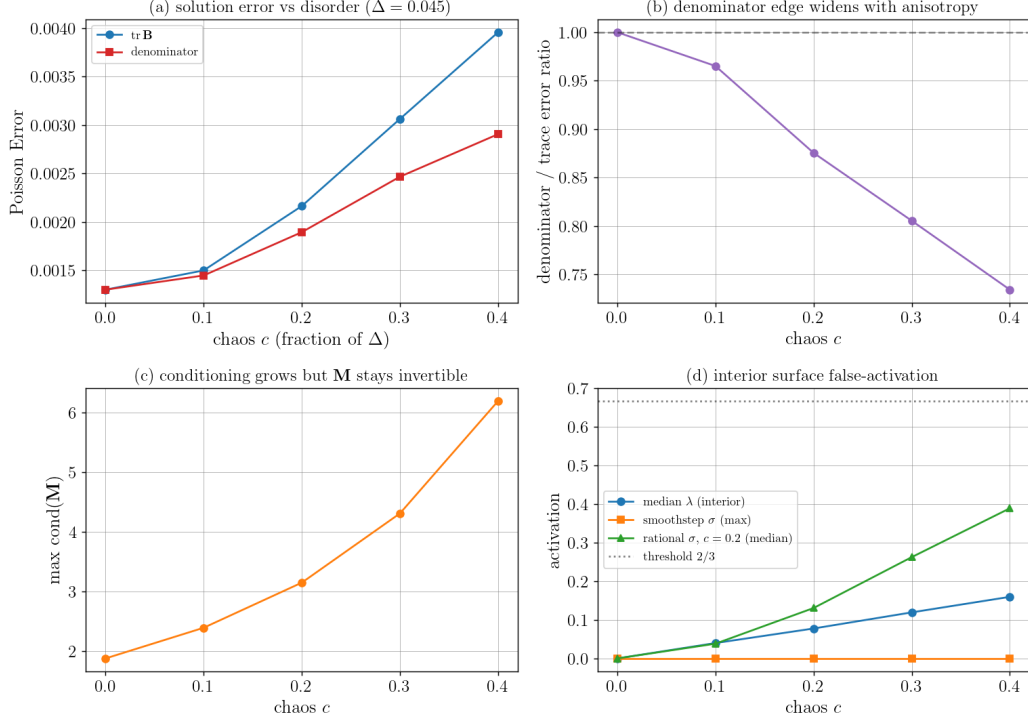


Figure 16: Disorder robustness of the Poisson solver showing the error, normalisation ratios, matrix condition numbers, and false surface activations.

residual divergence of the computed solenoidal part using Eq. (49):

$$\langle \nabla \cdot \mathbf{u} \rangle = \langle \nabla \cdot \mathbf{u}^* \rangle - \text{DG } \phi = \mathbf{L} \phi - \text{DG } \phi = (\mathbf{L} - \text{DG}) \phi. \quad (50)$$

The residual divergence is exactly the operator commutation defect $(\mathbf{L} - \text{DG})$ acting on the potential. The defect vanishes identically for staggered finite-difference or finite-volume discretisations. For those methods, we obtain $\mathbf{L} = \text{DG}$, making the decomposition exact. In this meshless case, we measure the non-adjointness of the operators. We must therefore answer two questions: is this residual small, and does it vanish under refinement?

We construct a manufactured Helmholtz-Hodge decomposition to provide an analytical ground truth. We define the stream function as $\psi = \sin \pi x \sin \pi y$ and the potential as $\phi^* = \cos 2\pi x \cos 2\pi y$. With these components, we build the field:

$$\mathbf{u}^* = \nabla^\perp \psi + \nabla \phi^*, \quad (51)$$

where we define the skew gradient as $\nabla^\perp \psi = (\partial_y \psi, -\partial_x \psi)$. This field has a known divergence, $\nabla \cdot \mathbf{u}^* = \nabla^2 \phi^*$. We also possess the analytical potential ϕ^* (up to a constant) and a divergence-free component $\nabla^\perp \psi$. This solenoidal component is tangential to the boundaries of the unit box because ψ vanishes along them. We prescribe the Neumann boundary condition on solid walls using $\nabla \phi^* \cdot \mathbf{n}$. This ensures the decomposition remains exact on any domain, including the wedge tank from Section 5. We feed the discrete divergence $\langle \nabla \cdot \mathbf{u}^* \rangle$ as the source term in Eq. (49), mimicking a real solver. This ensures that the identity in Eq. (50) holds discretely.

We confirmed the relation in Eq. (50) to round-off error. The directly measured divergence matches the commutation defect $(\mathbf{L} - \mathbf{DG})\phi$ to within $\sim 10^{-13}$ in the interior. A single projection step removes between 80 and 95% of the initial divergence. Furthermore, the residual exhibits convergence under refinement. The projection Neumann closure yields an order of 1.2 on a regular lattice, 1.0 on the wedge tank, and 0.4 on the box at 60% chaos. These rates mirror the supraconvergent solution orders reported in Section 5. The algebraic-ghost closure lowers the residual divergence on the box by about a third. It also significantly improves the accuracy of the potential, raising its convergence order from 0.9 to 1.7. However, the ghost closure leaves the convergence rate of the residual divergence at about 0.4. This is in contrast to

the substantial speedup brought to the Poisson solution in Section 5. The fact that an improved wall closure improves the potential but not the divergence convergence rate is telling. It indicates that the divergence-gradient pairing itself, rather than the Laplacian closure, limits the residual. The operator comparison makes this point precise. In Fig. 17(a), we plot the residual divergence against Δ for both closures on the box and the wedge tank. The decomposition is non-expansive. The solenoidal part carries less energy than the original field: $\|\mathbf{u}\|/\|\mathbf{u}^*\| \approx 0.47$. Moreover, the discrete inner product $\langle \mathbf{u}, \langle \nabla \phi \rangle \rangle$ between the solenoidal and potential components vanishes under refinement. This shows that the decomposition becomes orthogonal in the limit. Thus, we show that the renormalised operators are consistent for a single decomposition step. The residual divergence is exactly the non-adjointness defect, and this defect vanishes as $\Delta \rightarrow 0$.

We then isolated the choice of Laplacian. On an all-Dirichlet box, which every candidate operator can solve, we used the same renormalised gradient for $\mathbf{D}$ and $\mathbf{G}$. We then varied the Laplacian $\mathbf{L}$ in Eq. (49). We tested the uncorrected summation $N_i \sum_j W_{ij} f_{ij}$ (obtained by setting $w_{ij} = 1$ in Eq. (13)), the renormalised operator (with smooth-naive and sum normalisations), the GFDM operator from Section 5, and the Full Inverse operator of Asai et al. [8]. In Fig. 17(c), we report the residual. The uncorrected summation diverges. Its residual grows under refinement, which aligns with the well-known non-convergence of the uncorrected SPH Laplacian on irregular clouds [2, 4, 1]. This demonstrates that renormalisation is essential for projection consistency. In contrast, the renormalised, GFDM, and Full Inverse operators are all consistent and yield similar results. The two second-

order operators obtain only a marginal improvement in the residual compared to the linear-consistent renormalised operator. Equation (50) explains this. With a shared gradient, the gradient pairing $\mathbf{DG}$, rather than the accuracy of $\mathbf{L}$ itself, determines the residual $(\mathbf{L} - \mathbf{DG})\phi$. Consequently, a more accurate Laplacian alone cannot reduce the residual. We therefore show that projection consistency is a property of how we pair the divergence and the gradient, rather than the formal order of the Laplacian. This matches the central finding of this work. The boundary closure, and not the interior order, governs the final accuracy of the solution.

We can reduce the residual further by repeating the decomposition. We do this by solving Eq. (49) for the remaining divergence and applying a correction. This defect correction procedure lowers the residual to a floor of about one to three percent of the initial divergence. After that, the process stalls because the asymptotic contraction factor approaches unity. The iteration does not reach a completely divergence-free state. This is because the commutation defect has a near-null mode that we cannot remove, as shown in Fig. 17(b). If we feed the raw one-sided divergence at wall nodes directly to the source term rather than imposing it through the Neumann condition, the iteration diverges. This represents the flux-only Neumann limitation from Section 3 reappearing in the divergence field; we suggest that the divergence wall term is itself a candidate for algebraic-ghost stencil completion. Thus, the renormalised closure is best suited for a single Helmholtz-Hodge decomposition step. It yields a small residual divergence that vanishes under refinement, rather than a fully iterative path to a divergence-free field.

Helmholtz-Hodge decomposition with the renormalised FIRM operators

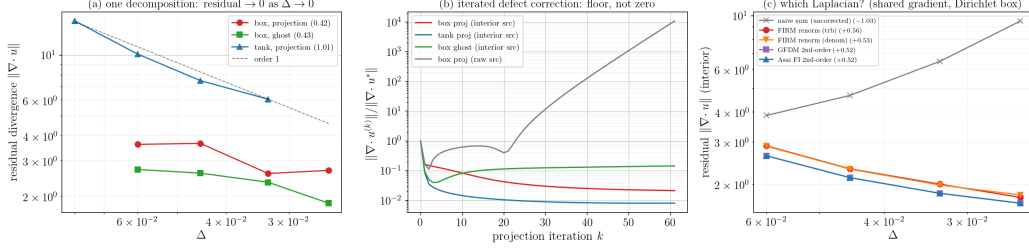


Figure 17: Divergence residual convergence, iterated defect correction, and operator choices in the Helmholtz-Hodge decomposition.

7. Discussion

7.1. Computational aspects

The closures only add local work during the interior assembly process. The Neumann projection uses a $d \times d$ projector built from at most K boundary normals. This requires a single $K \times K$ Gram matrix solve when we use the GGP method. The free-surface detection relies on the matrix-vector product $\mathbf{B}_i \mathbf{o}_{w,i}$, which we already compute during the standard assembly. The algebraic-ghost closure appends at most a few image samples for each Neumann node, re-using the same moment calculations. Thus, the boundary work is $O(1)$ per boundary node. We only incur it on the $O(N^{(d-1)/d})$ nodes near the boundaries. The assembled global system is sparse and non-symmetric because both the correction weights w_{ij} and the normalisations N_i are local to each row. A Krylov solver, such as BiCGStab or QMR with a Jacobi or ILU(0) preconditioner, is well-suited for solving this system, as discussed in [1]. We recommend the poly6 kernel at $h = 2.5\Delta$, or $h \in [2.0, 2.25]\Delta$ for quasi-uniform point clouds. For robust performance near corners, we recommend the GGP projection. Where high Neumann accu-

racy is required, we prefer the algebraic-ghost closure. The choices between AN and GGP, or between projection and ghosting, are isolated local code changes. All downstream calculations consume either the projected offset $\mathbf{o}_{w,i}$ or the completed row, regardless of the specific options chosen.

7.2. Operator and closure combinations

Every node corresponds to a single row in Eq. (22). Thus, we form a complete scheme by combining an interior Laplacian operator with a closure for each boundary type. In Table 2, we collect these combinations, showing their convergence orders and relative per-node costs. The diagonal closures, namely projection and Robin, require the renormalised operator. In contrast, we can pair the reflection ghosts (even reflection for Neumann flux and odd reflection for free surfaces) with any operator. We can also mix them with diagonal forms. For this reason, the rows in the table are representative rather than exhaustive. The smooth-naïve and sum normalisations of the renormalised operator yield the same convergence order when paired with diagonal closures. The sum normalisation only reduces the absolute error constant, so we list this diagonal scheme once. We only observe a difference when using even-ghost Neumann boundaries. There, the sum normalisation is exact for $|\mathbf{x}|^2$ and reaches about 1.8, whereas the smooth-naïve normalisation reaches a plateau at 1.4. This is why both formulations appear with the ghost closures.

Two rows in the table are not strictly instances of our unified row. We should clarify exceptions. First, the generalised finite difference operator does not fold boundary conditions into a single equation. It pins a Dirichlet node directly to its value, and a Neumann node carries a local Karush-

Table 2: Summary of numerical schemes showing their boundary closures, convergence rates, per-node costs, and divergence annihilation fractions.

Laplacian	Neumann	Free surface	order (int / Neu / surf)	cost	div. cleaned
smooth naive	projection	Robin	1.4/0.9/1.8	lowest	94%
smooth naive	ghost	ghost	1.4/1.4/2.0	low	94%
sum	ghost	ghost	1.4/1.8/2.0	low	94%
GFDM	constraint	ghost	1.8/0.9/2.0	high	94%
Full Inverse	ghost	ghost	1.8/1.8/2.0	moderate	94%

Kuhn-Tucker constraint block. Consequently, the global system mixes interior, pinned, and constrained row types. Second, the Full Inverse operator maintains a single row per node. However, we solve that row via a local least-squares system rather than the single renormalised sum of Eq. (22). Both methods are still free of physical boundary particles and extra global unknowns. The reflection ghost is ephemeral and we eliminate the constraint multiplier locally. Yet, neither method is a boundary-integrated, single-equation formulation. They are not instances of FIRM proper. We include them in the table as a second-order interior baseline (GFDM) and as a same-family second-order operator (Full Inverse) to show how we can transfer the algebraic-ghost closure. We define canonical FIRM as the renormalised operator with the boundary flux integrated directly into the operator. We place the Neumann flux on the right-hand side. We split the free-surface value, recast as the value-dependent Robin flux of Section 3.4, between the right-hand side and the diagonal of the same row. We achieve this without boundary nodes, constraint blocks, or auxiliary unknowns. This configuration corresponds to the first row of Table 2, which uses the smooth-naive

operator with projection and Robin closures. The ghost rows represent the stencil-completing variant. The name denotes this exactly: a flux-integrated renormalised meshless operator.

Table 2 delivers two practical messages. First, the renormalised operators introduced here are not the most accurate options available. A fully second-order interior operator, whether the GFDM least-squares operator or the Full Inverse operator of Asai et al. [8], attains a higher interior convergence order. Instead, the proposed construction offers convenience. The interior operator uses a single neighbour loop with no local linear solve. We integrate boundary conditions by type on a single code path with no auxiliary unknowns or tuning parameters. Furthermore, the reflection-ghost closure is operator-agnostic. This allows us to assemble a scheme simply by choosing one row and one closure per boundary type, rather than deriving a new operator from scratch. The cheapest combination in Table 2 is the smooth-naive operator with projection and Robin closures, which is also the simplest to implement.

The Helmholtz-Hodge decomposition reinforces this point. The efficiency of the divergence removal is essentially independent of the accuracy ordering. Every corrected scheme in Table 2, from the low-order renormalised operator to the second-order GFDM and Full Inverse operators, removes the same 94% of the initial divergence in a single step. This is because the divergence-gradient pairing, rather than the interior Laplacian order, sets the residual, as discussed in Section 6. Thus, the simple, low-order operator proves to be as effective as the more complex options, at a fraction of their cost. We can reduce the small remaining residual further by repeating the decomposition, driving the divergence down to a floor of about one percent. An

iterative application of the conveniently assembled low-order operator performs the Helmholtz-Hodge decomposition on par with the higher-accuracy alternatives.

8. Conclusions

We presented a boundary-integrated closure for renormalised meshless operators on elliptic boundary value problems. This method possesses three main features. First, we integrate boundary conditions by type on a single code path. A Neumann flux enters the right-hand side, and a Dirichlet or Robin value enters the diagonal. This allows us to assemble the interior, the Neumann boundary, the free surface, and the contact lines using a single kernel. Second, we impose the Neumann condition using a tangential projection in physical space before renormalisation. The general Gram projector treats arbitrary non-orthogonal corners exactly. A complementary algebraic-ghost closure completes the truncated boundary stencil without introducing additional unknowns, thereby removing the Neumann accuracy bottleneck. Third, the free surface is detected from the support-deficiency geometry of the cloud. It is enforced without tuning parameters or thresholds, as the enforcement strength cancels out to the total kernel weight.

The closures are exact for linear fields on any cloud, including truncated and corner rows, as verified to round-off error. Performance was tested on irregular-domain Poisson benchmarks for regular and highly irregular point clouds, and compared against a second-order GFDM baseline. These convergence results are reported in full. The renormalised interior operator is linear-consistent and supraconverges at an order of about 1.3 under strong

disorder. In those highly disordered regions, a fully second-order GFDM operator attains higher accuracy. The value closure supraconverges, whereas the flux-only Neumann closure acts as the main accuracy bottleneck. This is solved by the algebraic-ghost closure. It reduces the error in the Neumann region by about an order of magnitude, letting the Neumann region converge at the interior rate. With the sum normalisation, second-order convergence is approached at straight boundaries. The primary value of the proposed method lies in the generality, robustness, and parameter-free nature of the boundary closure. The method also benefits from convenient assembly. The interior operator uses a single neighbour loop with no local solves, and closures are selected by boundary type on one code path. Consequently, a complete scheme is assembled by combining existing parts rather than deriving a new operator from scratch, as shown in Table 2.

Several applications and extensions are foreseen for future research. The boundary closure is generic for elliptic boundary value problems on moving point clouds. A leading application is the Helmholtz-Hodge decomposition, which was examined in Section 6. In that context, the Neumann condition arises at solid boundaries, while the value condition is carried by the free surface. The pressure projection in incompressible flows is a key example. The operators are consistent for a single decomposition. The residual divergence is equal to their non-adjointness defect, which vanishes under refinement. The divergence-gradient pairing, rather than the Laplacian order, governs this residual. In practice, the simple, low-order operator cleans the same fraction of the divergence in a single step as the second-order GFDM and Full Inverse operators. An iterated decomposition reduces the resid-

ual further. Thus, the conveniently assembled low-order operator performs the Helmholtz-Hodge decomposition on par with the higher-accuracy alternatives. A full incompressible flow solver would also require a no-slip velocity Robin condition and a stress-free surface. These would inherit the same boundary-condition-by-type treatment, combined with a divergence-free time integration scheme, though these aspects were not assessed here. Incompressible free-surface hydrodynamics, including sloshing, dam break, and wave impact on moving point clouds, remains the motivating application. In those scenarios, the cloud-detected free surface represents the principal advantage because it requires no surface particles or threshold parameters. Another near-term application is the processing of scattered experimental velocity fields. This includes pressure reconstruction from particle image velocimetry. Here, a pressure Poisson problem is solved on a masked, irregular domain with Neumann data from the momentum balance at solid walls, and a value condition at the free surface or far field. The method can also be applied to the Helmholtz-Hodge filtering of noisy measured fields to recover their divergence-free component. Fluid-structure interaction and other moving-boundary problems are also candidates for this Lagrangian, geometry-detected treatment. A locally second-order-consistent interior operator, such as the Full Inverse renormalised operator of Asai et al. [8] which reinstates the cross-derivative coupling, lifts the overall accuracy. When paired with the reflection-ghost closure (using even reflection for Neumann walls, odd reflection for Dirichlet free surfaces, and composing them at contact lines), second-order accuracy is yielded throughout. This combination provides a second-order interior, a Neumann boundary at an order

of about 1.8, a detected curved free surface at about 2.0, and contact lines at the interior rate, all without adding extra global unknowns. The same reflection ghosts apply to the linear-consistent renormalised operator, where the odd-ghost free surface actually outperforms the diagonal-Robin value closure. A high-order flux-only Neumann treatment using the staggered KKT construction of Trask et al. [29] or the GFDM constraint row [27, 28] remains a topic for future study. Curvature-aware reflections, surface tension via a meshless curvature estimate, and three-dimensional validation are left for future work.

Acknowledgements

Placeholder.

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